DBMS: Data Design and Normalization

ASSIGNMENT 2.1

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Outline

1 MULTIPLE CHOICE QUESTIONS (MCQs)

MCQ Q1: FD Deduction – Question

Question

MCQ Q1 **Question:** Given $FD = \{AB \to C, CD \to E, E \to FG, G \to H\}$. Find which of the following can be a possible deduction from given functional dependencies.

- (a) $DFG \to E$
- (b) $ACD \to B$
- (c) $ABD \to E$
- (d) None of these

MCQ Q1: FD Deduction – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
To find a possible deduction, compute the closure of the left-hand side of each option. If the right-hand side attribute is in the closure, the FD is a valid deduction.

MCQ Q1: FD Deduction – Solution

Solution

Answer: (c) $ABD \rightarrow E$

For (a) $(DFG)^+ = DFGH$ (does not contain E)

For (b) $(ACD)^+ = ACDEFGH$ (does not contain B)

For (c) $(ABD)^+ = ABCDEFGH$ (contains E)

Therefore, $ABD \rightarrow E$ is a possible deduction.

MCQ Q2: 3NF with Singleton Keys – Question

Question

MCQ Q2 **Question:** Consider a relation R, all of its candidate keys are singletons. If R is in 3NF then R is _____.

- (a) BCNF
- (b) Not BCNF
- (c) May not be BCNF
- (d) Information is not sufficient

MCQ Q2: 3NF with Singleton Keys – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Analyze the conditions for 3NF and BCNF. For a relation in 3NF, every FD $X \to Y$ satisfies either (i) X is a superkey, or (ii) Y is a prime attribute. If all candidate keys are singletons, then any prime attribute is itself a candidate key.

MCQ Q2: 3NF with Singleton Keys – Solution

Solution

Answer: (a) BCNF

In 3NF, for any FD $X \rightarrow Y$, either X is a key or Y is prime. Since all keys are singletons, any prime attribute is a key. Therefore, if Y is prime, then Y is a key. This means Y determines all other attributes. This forces X to also be a key, satisfying the BCNF condition.

MCQ Q3: Minimal Cover – Question

Question

MCQ Q3 **Question:** Consider $R = ABCDEFGH$ and the following $FD's$: $H \to GD$, $E \to D$, $HD \to CE$, $BD \to A$. Identify the minimal cover of the given FD's?

- (a) $\{H \to G, E \to D, H \to C, HD \to E, BD \to A\}$
- (b) $\{H \to G, H \to D, E \to D, H \to C, BD \to A\}$
- (c) $\{H \to G, E \to D, H \to C, HD \to C, BD \to A\}$
- (d) $\{H \to G, E \to D, H \to C, H \to E, BD \to A\}$

MCQ Q3: Minimal Cover – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Find the minimal cover by:

1. Right Reduction: Decompose FDs so each has a single attribute on the right.
2. Left Reduction: Minimize the left-hand side of each FD.
3. Remove Redundant FDs.

MCQ Q3: Minimal Cover – Solution

Solution

Answer: (d)

Step 1: Right reduce:

$H \to G, H \to D, E \to D, HD \to C, HD \to E, BD \to A$

Step 2: Left reduce $HD \to C$ and $HD \to E$: Both cannot be reduced to $H \to C$ or $H \to E$ as H^+ does not contain C or E.

Step 3: Remove redundancies: $HD \to E$ is redundant (can be derived). $HD \to C$ is also redundant.

Minimal cover: $\{H \to G, E \to D, H \to C, H \to E, BD \to A\}$

MCQ Q4: Normal Form Identification – Question

Question

MCQ Q4 **Question:** Consider a relation $R(A, B, C, D, E)$ with sets of $FD's$ $\{A \to B, BC \to E, ED \to A\}$. Given R is in which highest normal forms?

- label=(a) 1NF
- label=(b) 2NF
- label=(c) 3NF
- label=(d) BCNF

MCQ Q4: Normal Form Identification – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Find all candidate keys. Check each FD against BCNF and 3NF conditions.

MCQ Q4: Normal Form Identification – Solution

Solution

Answer: (c) 3NF

Candidate Keys: ACD, BCD, EDC. Prime attributes: A, B, C, D, E.
Check BCNF: $A \to B$ (A not a key), $BC \to E$ (BC not a key), $ED \to A$ (ED not a key) - All violate BCNF.
Check 3NF: $A \to B$ (B is prime), $BC \to E$ (E is prime), $ED \to A$ (A is prime) - All hold.
Therefore, highest normal form is 3NF.

MCQ Q5: Lossless Join Decomposition – Question

Question MCQ Q5

Question: Let R be the relational schema with FD's F, is decomposed into R1 and R2 is lossless join decomposition only if **Options:**

(a) (i) R1 R2 R (ii) R1 R2 (iii) R1 R2 gives candidate key that should be primary key in either R1 or R2

(b) (i) R1 R2 = R (ii) R1 R2 (iii) R1 R2 gives candidate key that should be primary key in either R1 or R2

(c) (i) R1 R2 = R (ii) R1 R2 = (iii) R1 R2 gives candidate key that should be primary key in either R1 or R2

(d) (i) R1 R2 = R (ii) R1 R2 (iii) R1 R2 gives superkey of at least one of R1 or R2

MCQ Q5: Lossless Join Decomposition – Strategy

Problem-Solving Strategy

- Recall the lossless join decomposition theorem.
- For a decomposition of R into R1 and R2 to be lossless, the intersection of R1 and R2 must be a superkey of at least one of R1 or R2.
- The union of R1 and R2 must equal R.

MCQ Q5: Lossless Join Decomposition – Solution

Solution

Answer: (b)

Option (b) correctly states all three conditions: (i) R1 R2 = R, (ii) R1 R2 , (iii) R1 R2 gives candidate key that should be primary key in either R1 or R2.

MCQ Q6: Normal Form of Relation S – Question

Question

MCQ Q6 **Question:** Consider the relation schema $S = \{A, B, C, D\}$ and the following functional dependencies on S: $A \to BCD$, $B \to C$, $CD \to A$ Which of the following is true?

- (a) S is in 3NF and also in BCNF
- (b) S is in 3NF but not in BCNF
- (c) S is in 2NF but not in 3NF
- (d) S is in BCNF but not in 3NF

MCQ Q6: Normal Form of Relation S – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Find candidate keys. Check each FD against BCNF and 3NF conditions.

MCQ Q6: Normal Form of Relation S – Solution

Solution

Answer: (b) S is in 3NF but not in BCNF

Candidate keys: A, CD. Prime attributes: A, C, D.

Check BCNF: $A \to BCD$ (A is key - holds), $B \to C$ (B not a key - violates), $CD \to A$ (CD is key - holds).

Check 3NF: $B \to C$ (C is prime - holds).

Therefore, relation is in 3NF but not BCNF.

MCQ Q7: Two-Attribute Relation – Question

Question

MCQ Q7 **Question:** A relation R with two attributes is always in

- 1NF
- 2NF
- 3NF
- BCNF

MCQ Q7: Two-Attribute Relation – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Analyze all possible functional dependencies for a two-attribute relation.

MCQ Q7: Two-Attribute Relation – Solution

Solution

Answer: (d) BCNF

Possible FDs: $A \to B$ (A is key), $B \to A$ (B is key), or $AB \to AB$ (AB is key), or no non-trivial FDs. In all cases, every FD's left-hand side is a superkey. Therefore, the relation is always in BCNF.

MCQ Q8: Decomposition Properties – Question

Question

MCQ Q8 **Question:** Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of FD, $F = \{AB \to C, A \to DE, B \to F, F \to GH, D \to IJ\}$. R is decomposed into $R_1 = \{A, B, C, D, E\}$, $R_2 = \{B, F, G, H\}$, $R_3 = \{D, I, J\}$. Then the decomposition of R is

- (a) Lossless join and dependency preservation
- (b) Lossy but dependency preservation
- (c) Lossless join but not dependency preservation
- (d) Lossy and loss of dependency

MCQ Q8: Decomposition Properties – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Check for lossless join by finding common attributes between decomposed relations and checking if they form a superkey in one of the relations.
Check dependency preservation by verifying if the union of FDs projected on each relation is equivalent to the original FD set.

MCQ Q8: Decomposition Properties – Solution

Solution

Answer: (a) Lossless join and dependency preservation

Lossless Join: Intersection of R1 and R2: B. $B^+ = BFGH$, so B is a key in R2. Intersection of (R1 $\bowtie$ R2) and R3: D. $D^+ = DIJ$, so D is a key in R3. Therefore, decomposition is lossless.

Dependency Preservation: Projections: R1: $A \to DE$, $AB \to C$; R2: $B \to F$, $F \to GH$; R3: $D \to IJ$. Union of projections is equivalent to original F. Therefore, dependency preservation holds.

MCQ Q9: Determining Normal Form – Question

Question MCQ Q9

Question: Consider the following relation
Vehicle (vehicle No, sold_date, salesmanNo, commission%, discount_amt)
Assume that a vehicle may be sold by multiple salesman and vehicle (vehicle No, salesmanNo) is Primary key. Additional dependencies are Sold_date $\rightarrow$ discount_amt and salesmanNo $\rightarrow$ commission%.
Given relation is in which normal form? **Options:**

(a) 1NF

(b) 2NF

(c) 3NF

(d) None of these

MCQ Q9: Determining Normal Form – Strategy

Problem-Solving Strategy

- Identify the primary key and check for partial and transitive dependencies.
- Verify if the relation satisfies the basic requirements of 1NF.
- In 1NF, all attributes must have atomic (indivisible) values.

MCQ Q9: Determining Normal Form – Solution

Solution

Answer: (d) None of these

Primary key: (vehicle No, salesmanNo)

FDs given: Sold_date $\rightarrow$ discount_amt, salesmanNo $\rightarrow$ commission%

The issue is that commission% contains a special symbol (%), which violates the atomic value requirement of 1NF. The percentage symbol makes this attribute non-atomic. Therefore, the relation is not even in 1NF.

Hence, the correct answer is (d) None of these.

MCQ Q10: GBN Window Size for Maximum Efficiency – Question

Question MCQ Q10

Question: A 3000 km communication line uses Go-Back-N protocol. Propagation speed is 6 ms/km, data rate is 1.544 Mbps, and frame size is 64 bytes. Acknowledgment processing time is negligible. What sender window size is needed to achieve 100% efficiency? **Options:**

(a) 32

(b) 63

(c) 110

(d) 219

MCQ Q10: GBN Window Size for Maximum Efficiency – Strategy

Problem-Solving Strategy

- For 100% efficiency: Window size = (2 × Propagation delay + Transmission time) ÷ Transmission time.
- Transmission time = Frame size (bits) ÷ Data rate.
- Propagation delay = Distance × Propagation speed.
- Pay careful attention to units (ms vs seconds).

MCQ Q10: GBN Window Size for Maximum Efficiency – Solution

Solution

Answer: (c) 110

Frame size $= 64$ bytes $= 64 \times 8 = 512$ bits

Transmission time $= 512 \div (1.544 \times 10) = 0.000332$ sec $= 0.332$ ms

Propagation delay $= 3000$ km $\times$ 6 ms/km $= 18{,}000$ ms $= 18$ sec

Window size $= (2 \times 18 + 0.000332) \div 0.000332 \quad 108.4 + 1 \quad 110$

MCQ Q11: Candidate Key Identification – Question

Question

MCQ Q11 **Question:** Given the relational schema $R(A, B, C, D, E)$ with the following functional dependencies: $A \to B$, $BD \to E$, $C \to AD$, $E \to C$ Find which of the following is not a candidate key?

- label=(a) E
- label=(b) C
- label=(c) BD
- label=(d) A

MCQ Q11: Candidate Key Identification – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Compute the closure of each option and check if it determines all attributes.

MCQ Q11: Candidate Key Identification – Solution

Solution

Answer: (d) A

$E^+ = ECADB = ABCDE \rightarrow$ Candidate key

$C^+ = CADBE = ABCDE \rightarrow$ Candidate key

$(BD)^+ = BDECA = ABCDE \rightarrow$ Candidate key

$A^+ = AB$ (does not contain C, D, E) $\rightarrow$ Not a candidate key

MCQ Q12: Minimal Cover Size – Question

Question

MCQ Q12 **Question:** Find the number of $FD's$ in the minimal cover of following FD set
$FD = \{A \to B, CD \to A, CB \to D, CE \to D, AE \to F, AC \to D\}$.

- (a) 2
- (b) 3
- (c) 4
- (d) 5

MCQ Q12: Minimal Cover Size – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Find the minimal cover by removing redundant FDs and reducing left/right sides.

MCQ Q12: Minimal Cover Size – Solution

Solution

Answer: (d) 5

Check redundancy: $AC \to D$ can be derived from $A \to B$ and $CB \to D$: $A \to B \Rightarrow AC \to BC$ and $BC \to D$ (but we have $CB \to D$, which is the same). Therefore, $AC \to D$ is redundant.

All other FDs are non-redundant and minimal.

Minimal cover: $\{A \to B, CD \to A, CB \to D, CE \to D, AE \to F\} \to 5$ FDs.

MCQ Q13: Primary Key Choices – Question

Question

MCQ Q13 **Question:** Find the number of possibilities to choose a primary key of the relation $R(A, B, C, D, E)$.

- (a) 5
- (b) 25
- (c) 31
- (d) 120

MCQ Q13: Primary Key Choices – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
A primary key can be any non-empty subset of attributes. Count all non-empty subsets.

MCQ Q13: Primary Key Choices – Solution

Solution

Answer: (c) 31

Number of non-empty subsets of a set of 5 elements:
$2^5 - 1 = 32 - 1 = 31$.

MCQ Q14: Candidate Key Count – Question

Question

MCQ Q14 **Question:** Let R is a relation schema, $R(A, B, C, D)$ and $F = \{A \to B, B \to C, C \to A\}$ is the set of functional dependency. How many candidate keys are there?

- label=(a) 1
- label=(b) 2
- label=(c) 3
- label=(d) None of these

MCQ Q14: Candidate Key Count – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Find all attributes that can determine all other attributes. Note that A, B, and C form an equivalence class.

MCQ Q14: Candidate Key Count – Solution

Solution

Answer: (c) 3

From $A \to B$ and $B \to C$, we get $A \to C$. Also $C \to A$, so A, B, C are equivalent. $A^+ = ABC$ (D not included). To determine D, we need D itself. So candidate keys are AD, BD, CD. Therefore, 3 candidate keys.

MCQ Q15: MVD Implied Tuples – Question

Question

MCQ Q15 **Question:** Suppose we have a relation $R(A, B, C)$ with multivalued dependency $\{A \twoheadrightarrow B\}$. If we know that tuples $(a, b1, c1), (a, b2, c2), (a, b2, c3), (a, b3, c3)$ are in R what other tuples do we know must also be in R?

- (a) $(a, b1, c2), (a, b1, c3), (a, b2, c1)$
- (b) $(a, b1, c2), (a, b2, c1), (a, b3, c1), (a, b3, c2)$
- (c) $(a, b1, c2), (a, b1, c3), (a, b2, c1), (a, b3, c1), (a, b3, c2)$
- (d) None of the above

MCQ Q15: MVD Implied Tuples – Strategy

Problem-Solving Strategy

Problem Solving Strategy:
Apply the MVD rule: If $A \twoheadrightarrow B$, then for a fixed A value, the set of B values is independent of C. This means all combinations of B and C values that appear with that A must be present.

MCQ Q15: MVD Implied Tuples – Solution

Solution

Answer: (c)

With A = a, we have B values: $\{b1, b2, b3\}$ and C values: $\{c1, c2, c3\}$.
All combinations must be present: a $\times$ $\{b1, b2, b3\}$ $\times$ $\{c1, c2, c3\}$.
Missing tuples: $(a, b1, c2)$, $(a, b1, c3)$, $(a, b2, c1)$, $(a, b3, c1)$, $(a, b3, c2)$.